

Aarav Jayswal

Tampa, Florida

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TECHNICAL SKILLS AND COURSES

Programming Languages: C, Python, MATLAB, Arduino

Platforms/Tools: KiCAD, Vivado, Verilog, Pspice, AutoCAD, ADS, Oscilloscope, Spot Welding

Courses: Electronic materials, Electrical System II, Electromechanical Systems, Power Lab, Wireless System Design Lab, RF I.

PROJECTS

PID Temperature Controller & Custom Power Supply(Buck Converter)

PSpice, Timer, C, PID Control

- Implemented a closed-loop thermal management system management system on an Arduino, driving a DC fan via PWM based on continuous PID calculations from DHT11 feedback.
- Integrated a custom 10V-40V to 5V buck converter to power the logic circuitry, utilizing an LM2576 and LC filter to minimize output voltage ripple.

Temperature Monitoring System

Arduino, RGB Led, Tinker Cad

- Developed an Arduino-based temperature monitoring system using a 10K potentiometer as an analog temperature input, with real-time temperature indication through an RGB LED.
- Implemented inverted PWM control for the common-anode RGB LED and programmed a high-temperature alarm using a red LED that flashes when the analog reading reaches the set threshold.

Line Following Robot – Hardware Lead

Arduino, C Programming, Analog to PWM conversion, I²C, IR sensors, Motor Driver, Tinker Cad

- Used IR sensors to measure the reflection of infrared radiation and implemented I2C communication protocol to display the speed and name of the model.
- Utilized Arduino to interpret IR sensor signal and send PWM signal by analyzing the potentiometer value to control the movement through a motor driver for responsive operation.

EXPERIENCE

Engineering Intern – Orbital Corporation of Tampa

May 2026 – Aug 2026

Wire harness, Cable assembly, Blueprints

- Performed Cable assemblies and wiring Harness per the drawing/blueprints.
- Assembling Headsets and Handsets per drawing and performing testing on all the assemblies.

EDUCATION

University of South Florida, Tampa, FL

August 2024 – May 2028

Major: Bachelor of Science in Electrical and Electronics Engineering

GPA: 3.87/ 4.00

Awards: Green and Presidential merit based Scholarship

Dean's list : Spring 2025, Fall 2025, Spring 2026

INVOLVEMENTS

IEEE Electrathon Team, University of South Florida

Aug 2025 – May 2026

- Designed and assembled the battery pack, integrating it with a Battery Management System (BMS) to provide a reliable power supply for the vehicle.

Software Shadow – Association of Computing Machinery (USF)

Aug 2024 – May 2025

- Led weekly project hubs for 30+ participants and conducted beginner-to-intermediate Python workshops.